

Carbon emissions in China's cement industry: A sector and policy analysis



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ABSTRACT

This paper analyzed the changes in carbon-dioxide (CO₂) emission in China's cement industry over the period 1991–2010. Logarithmic Mean Divisia Index (LMDI) was applied for the decomposition analysis. Five determinants were examined and the results divided into five time intervals, according to China's five-year economic plans. The results show that labor productivity was the major driving force of increase in CO₂ emission in China's cement industry, and energy intensity was the major contributor to CO₂ emission reduction. The other three factors also generated certain level of effects. This article provides some policy suggestions which could be adopted to reduce CO₂ emission in this energy-intensive industry.

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1. Introduction

Climate change is one of the most dominant topics in recent years, and significant efforts and researches have been focused on

this issue. Despite global acclaimed efforts on emission reduction, fossil fuel used in all aspects of the economy is still at a high level and threatens the earth and human beings. Due to the globalization trend in the past decades, pollution-intensive industries were transferred to developing countries, including China. China had always been ranked at the top of the list of the world's largest emitter. Its carbon emission figure in 1990 was 2.24 billion tons, but the country witnessed a substantial 400% increase in the following 20 years, amounting to 8.97 billion tons of carbon emission

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in year 2011 (accounted for 26.4% of the world's total emission according to BP [1]).

Due to both domestic and international pressure, the Chinese government had taken various actions to reduce its emissions. A series of policies and regulations has been initiated and published before the 2009 Copenhagen conference. The major emission reduction target has been set at reducing of energy consumption per unit of GDP by 40–45% in 2020 compared to the 2005 level. To achieve this objective, the central government had assigned specific emission reduction goals to each administration district and to each industrial sector. However, due to the unique characteristics of the different industries, it is necessary to comprehensively analyze the situation in high energy-consuming industries.

Currently, China produces the largest amount of cement in the world. According to statistics from CMIIT [2], China produced 1.87 billion metric tons of cement in 2010, accounting for around 57% of global cement production [3]. The average annual growth rate was 11.6%, which accumulated to 790% growth rate over the past two decades. Meanwhile, the enormous scale of fixed assets investments and infrastructure constructions had made China the largest cement consumer in the world. Cement production is one of the most energy-intensive industries. As can be seen from Fig. 1, China's cement industry consumed a total of 4542 PJ (155 Mtce) of energy in year 2009, which is about 7.1% of China's total final energy use and 10.1% [4] of the final energy consumption of the industrial sector [5].

Limestone calcinations and heavy fossil fuel combustion during cement production process also lead to extensive CO₂ emissions in China. The amount of CO₂ emissions from China's cement industry was 1073 Mt in 2009, and accounts for about 15% [6] of China's total greenhouse gas (GHG) emissions [7] and over 80% of CO₂ emissions from the construction and building sector, as shown in Fig. 2. According to Xu et al. [5], the cement industry could be one of the primary emission sources, which accounts for about 30% of total industrial particulate matter emissions nationwide in year 2009 [8]. Due to its role in global energy conservation and CO₂ mitigation, the cement industry had drawn huge attention globally. In 2009, IEA and WBCSD (World Business Council for Sustainable Development) published the first single industrial technology roadmap "Cement Technology Roadmap to 2050" [9]. Several industrial energy-conservation objectives were set in the roadmap, and Xu et al. [10] made a comparative analysis of the possibility of China's cement industry to meet the targets by potential technological advances.

Therefore, this paper will adopt the Logarithmic mean weight Divisia Index (LMDI) to decompose the factors driving carbon emission in China's cement industry. The decomposition will help policy makers and industrial professionals to know the degree of

China's Total GHG Emission

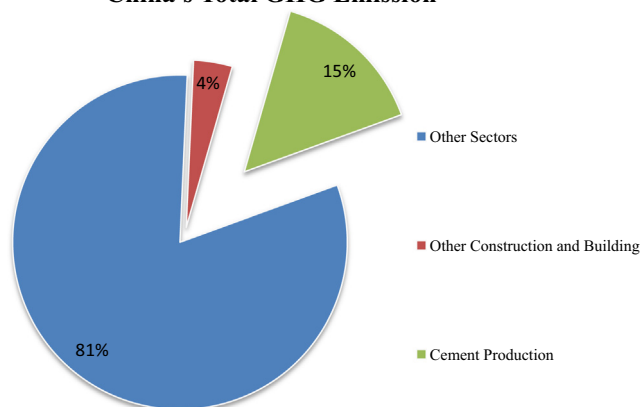


Fig. 2. China's total GHG emission.

the importance of each factor, and then strategize and act accordingly. The next part of this paper will review the literatures related to factor decomposition and studies on emission issues in the cement industry. The third part will analyze the status quo of carbon emission in China's cement industry. The fourth section will introduce the methods and methodology used in this paper. The fifth part will be the decomposition analysis, results and discussions. The last section of this paper contains the recommendations and conclusions according to the above analyses.

2. Literature review

Past studies on related factors of CO₂ emissions using quantitative analysis of the contribution rate of each factor can be categorized into two: econometric analysis based on the Environmental Kuznets Curve (EKC), and decomposition analysis based on the identical equations of carbon emissions. The underlying theory of the EKC method is that the quality of the environment has a reverse-U shape relationship with income per capita. Related literatures used national or regional income per capita data to explain the relationship with carbon emissions. Additional factors such as industrial structure, urbanization rate, technology level and trade openness had also been taken into accounts in some measurement models to examine the existence of EKC (Fried and Getzner [11]; Richmond and Kaufmann [12]; Lantz and Feng [13]; Wagner [14]; Auffhammer and Carson [15]; Ang [16]). However, for a single industry, the carbon emissions are additional products of its own production activities, and income per capita should not be a proper explaining variable as it does not have direct impacts. Meanwhile, the omission of other variables would generate some endogenous problems. Therefore, to study carbon emission of the cement industry, this paper will adopt the decomposition method.

The conventional decomposition studies on carbon emissions were based on identical equation of KAYA. The main data were aggregated at the national or industrial level. After the decomposition, the general contribution factors are: Energy Intensity (EInt), Energy Structure (EStr), Industrial Scale (ISca). Others studies had also expanded the KAYA equation to obtain the effects of such variables as urbanization rate (UrbR) and so on. For this, structural decomposition and index decomposition are the two most popular methods. Ang et al. [17] presented the so called Log-Mean Divisia Index (LMDI). This method produces decomposition results with no residual error and with higher degree of decomposition accuracy. Therefore, LMDI had been largely applied in studies of China's carbon emission.

China's Total Final Energy Consumption

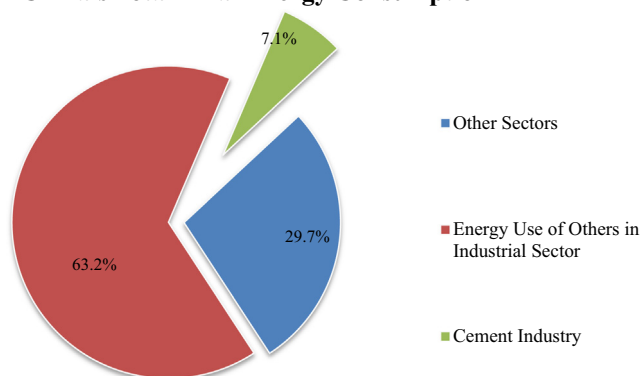


Fig. 1. China's total final energy consumption.

Liu et al. [18] analyzed the change in industrial carbon emissions from 36 industrial sectors in China over the period 1998–2005, based on the time series decomposition of the Log-Mean Divisia Index (LMDI). Fan et al. [19] quantified the driving force of China's primary energy-related carbon intensity for the period 1980–2003, and found that reduction in real energy intensity was the biggest contributor to the decline in carbon intensity. Zhang et al. [20] analyzed the nature of the four factors (CO_2 intensity, energy intensity, structural changes and economic activity) of CO_2 emission in China for the period 1991–2006. The results showed that economic activity has the largest positive effect on CO_2 emission changes.

Apart from studies that analyze carbon emission at the national aggregate level, there are other studies that focused on the carbon emission of specific industrial sectors. Sun et al. [21] decomposed total change in CO_2 emissions in China's iron and steel industry into four factors: emission factor effect, energy structure effect, energy consumption effect, and steel production effect. The results showed that steel production effect was the major factor responsible for the rise in CO_2 emissions; whereas energy consumption effect contributed most to the reduction in CO_2 emissions. Lin and Wang [22] presented another analysis in CO_2 emissions reduction of China's iron and steel industry by considering its industrial CO_2 emissions and anticipating the resulting reduction potential. Tian et al. [23] addressed the energy-related GHG emission trajectories, features, and driving forces in Chinese ISI for 2001–2010. The results showed that production scale effect is the main driving factor for energy related GHG emission increase in China's ISI, while energy intensity effect and emission factor change effect offset the total increase and energy structure has marginal effect. Moreover, this method was also applied in studies of CO_2 emissions in other Chinese industrial sectors, such as textile industry [24], electricity generation [25] and the whole industrial sector [26].

For the cement industry, many studies, mostly from the perspective of technological improvement potentials, have suggested increasing energy efficiency. Liu et al. [27] analyzed the cost and obstacles of adopting new cement production technologies. Some other studies estimated the energy mitigation potentials from an engineering aspects (Li [28]; Lei et al. [29]; Worrell et al. [30]; Zeng [31,32]). Moreover, Hasanbeigi et al. [33] completed a comparison of Chinese cement companies with international practices. Cui and Liu [34] explicated some energy saving measures in limestone calcinations, and fossil fuel combustions. Lei et al. [29] proposed a method for improving cement industrial energy efficiency and reducing CO_2 emissions by renovating outdated shaft kilns. Price et al. [35] compared international energy policies. Xu et al. [5] analyzed the change in energy consumption and CO_2 emissions in China's cement industry and its driving factors over the period 1990–2009 by applying a log-mean Divisia index (LMDI) method, and found that the growth of cement output is the most important factor driving increase in energy consumption.

Wang et al. [36] presents a GHG emission inventory of the cement industry in China and found that the main driving factors that influence changes in GHG emissions in the cement industry was cement production activity effect and clinker production activity effect. Teller et al. [37] verifies that 95% of CO_2 emissions in cement life cycle were released in the production process, while only 5% were released from the transportation stage.

The main contribution of this paper is not methodology, but an important and relevant sector analysis and policy research. In fact, in empirical research, we often study the same topic with same methodology for different countries, given that the topic has a deserved importance and methodology is a correct one. The CO_2 emission of China's cement industry is larger than the total CO_2 emissions of many countries. Therefore, a good sector research,

though with limited contribution to methodology, is also important in the course of sustainable development.

The cement industry is one of the largest CO_2 emitting industrial sectors both China-wide and worldwide. World cement production in 2011 reached 3.6 billion tonnes whilst produced over 2 billion tonnes of CO_2 . And China is now the No. 1 cement consumer and producer in the world. With its domestic needs of urbanization and infrastructure construction, 58% of global cement output in 2012 was produced by China. China's cement industry consumes more than 7% of the nation's total final energy, and generates more than 15% of total GHG emission. With air pollution becoming one of the most urgent public concerns in today's China, the cement industry as one of the primary sources of emissions, has attracted a lot of attention in term of energy and environmental policy.

Further, with size of the industry and related CO_2 emissions, a good understanding of China's CO_2 emissions of the cement industry and effective policy measures based on the study, are very important for national and global efforts in CO_2 reduction. This study will also provide some useful analysis and policy recommendations for other developing countries, as their cement industries are actually in the path of development similar to China. This is also a very important contribution to the body of sectoral CO_2 reduction literature.

There are substantial amount of research in CO_2 emissions in China, but very few studies on the cement industry. Therefore, for the importance of this industrial sector in both China and worldwide, we believe this paper will sharpen the policy effort in CO_2 emission reduction in China, and contribute to the general effort of related policy discussions of energy conservation and CO_2 mitigation, and sustainable development.

3. Overview of China's cement industry

China's cement industry had seen a rapid growth over the last two decades, along with the huge economic boost. As can be seen from Fig. 3, cement output soared from 61 billion RMB in 1991 to 571 billion RMB in 2010. While the output experienced a 936% growth, the proportion of the cement industry in China's GDP had been on a declining trend from about 2.75% in 1991 to less than 1.5% in 2010. This matches the economic transition policies of the government since the beginning of the administration of President Jiang Zemin.

The energy consumption structure in China's cement industry is relatively stable. Using mostly fossil fuels and electricity, the proportion of the energy sources has remained at a certain level. As can be seen from Fig. 4, the ratio of electricity and fossil fuel consumption had remained around 25–30% over these two decades.

Fig. 5 shows that, year 2000 was the turning point of the energy intensity of the cement industry. Technological

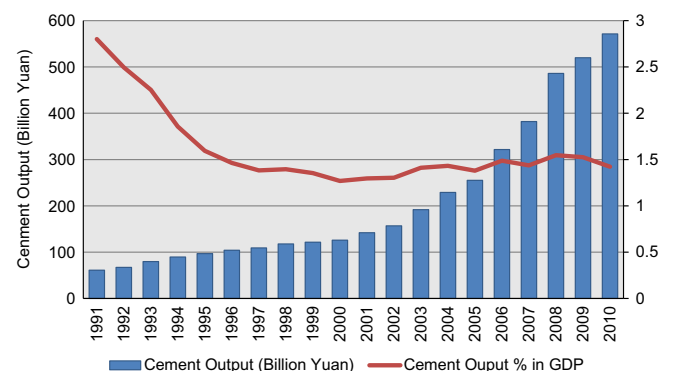


Fig. 3. Output of China's cement industry.

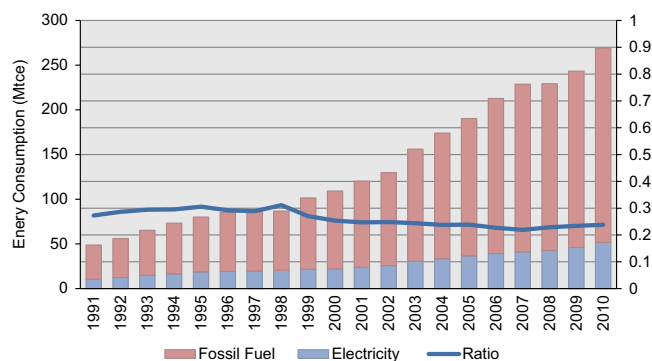


Fig. 4. Energy consumption structure in Cement industry.

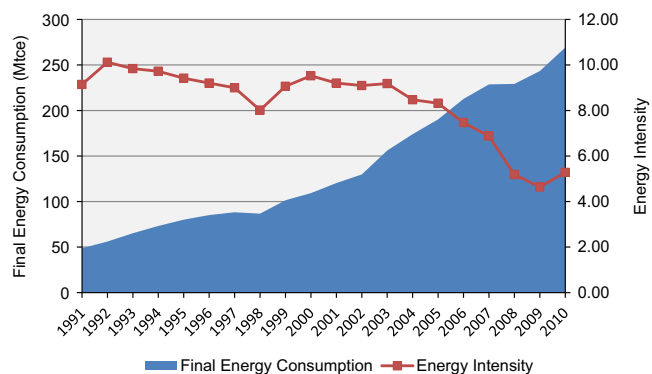


Fig. 5. Final energy consumption and energy intensity in China's cement industry.

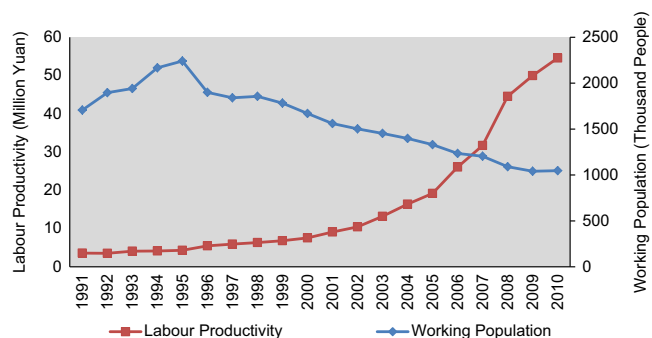


Fig. 6. Labor productivity and working population.

improvements and huge scale of new type kiln adoption have huge impacts on energy conservation and CO₂ emission mitigation in China's cement industry. This technological advancement also resulted in improving labor productivity in the cement industry, as shown in Fig. 6. Although labor productivity has long history of upward trends, it had experienced a six-fold increase since the beginning of 2001. At the same time, the total working population in the cement industry fell from the peak of about 2.3 million people to just over 1 million in year 2010.

Alongside increasing production in the cement industry, CO₂ emissions had been boosted as a result of this industrial expansion rate. As shown in Fig. 7, in the first 14 years, CO₂ emissions had almost the same growth rate as cement output until 2005. Due to high technological renovation and mechanical innovation, the gap between cement output growth rate and CO₂ emission growth rate started to increase. Since then, even though China's cement production had shown a rapidly increasing trend, the increasing pace of CO₂ emission had started to slow down.

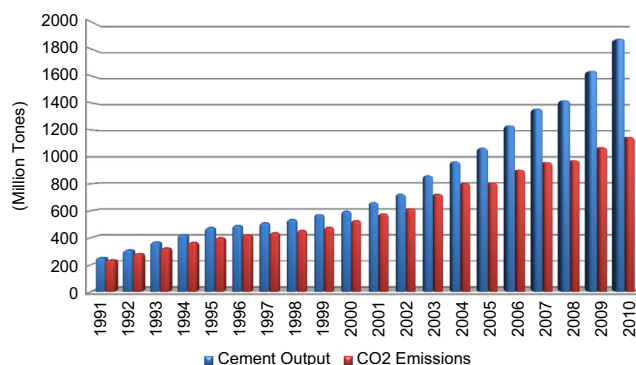


Fig. 7. Cement output vs. CO₂ emissions.

4. Methodology and data sources

4.1. Methodology

For the purpose of identifying the affecting factors of energy consumption and related CO₂ emission in China's cement industry, we employed the Logarithmic Mean Divisia Index (LMDI) method. As mentioned earlier, this method has sound theoretical foundation, adaptability and convenience.

The Kaya identity divides a country's CO₂ emission into carbon intensity, energy intensity, income per capita and population. Considering the particularity of the industry's CO₂ emission, we modified the Kaya identity to fit the characteristics of China's cement industry. The driving forces which influence the change in CO₂ emission include the following five factors:

Carbon intensity effect: It is the ratio of CO₂ emissions over fossil fuel consumption. It changes because the proportion of each kind of fossil fuel changes over time and the emission coefficients of different kinds of fuels are different.

Energy structure effect: Its value represents the share of fossil fuel in the whole energy consumption. Because we don't take emission from electricity and heat consumption into account, the change in the share of fossil fuel will affect the total emission.

Energy intensity effect: It is formulated as the ratio of energy consumed over industrial output. The change in the amount of energy per unit of output reflects the impact of technology improvement.

Labor productivity effect: It represents the outputs produced by each employee in the industry.

Industry scale effect: The number of employees in this industry has direct impact on energy use.

In this paper, the index decomposition analysis of CO₂ emissions in China's cement industry is conducted by the following equation:

$$C = \frac{C}{E_f} \cdot \frac{E_f}{E} \cdot \frac{E}{Y} \cdot \frac{Y}{w} \cdot w \quad (1)$$

The meanings of the parameters in Eq. (1) can be found in the first part of Table 1. Where C represents the total CO₂ emission of China's cement industry; E_f represents the proportion of fossil fuel in the cement industry's total energy consumption; E represents total energy consumption, Y represents industrial added value of China's cement industry; w is the total working population of China's cement industry. Also, the multipliers in Eq. (1) represent different effects and the exact meaning and abbreviation of each multiplier is shown in the second part of Table 1.

Rewriting Eq. (1), we obtain:

$$C = CI \times ES \times EI \times LP \times IS \quad (2)$$

Using the logarithmic mean Divisia index method, we divide the change in CO₂ emission into five parts. The aggregated change

in CO₂ emissions recorded in year *t* in comparison with its level in a base year is calculated as follows:

$$\Delta C = C_t - C_0 = \Delta C_{CI} + \Delta C_{ES} + \Delta C_{EI} + \Delta C_{LP} + \Delta C_{IS} \quad (3)$$

These components are written as:

$$\Delta C_{CI} = L(C_0, C_t) \cdot \ln(CI_t/CI_0), \quad \Delta C_{ES} = L(C_0, C_t) \cdot \ln(ES_t/ES_0)$$

$$\Delta C_{EI} = L(C_0, C_t) \cdot \ln(EI_t/EI_0), \quad \Delta C_{LP} = L(C_0, C_t) \cdot \ln(LP_t/LP_0)$$

$$\Delta C_{IS} = L(C_0, C_t) \cdot \ln(IS_t/IS_0)$$

where $L(C_0, C_t) = \frac{C_t - C_0}{\ln(C_t/C_0)}$, is called logarithmic mean weight.

4.2. Data

This paper is based on data from China Statistical Yearbooks, China Energy Statistical Yearbooks, and CEIC database. The period from 1991 to 2010 was chosen for the decomposition using LMDI method. The definition of the target industry in this paper is given by China's National Bureau of Statistics in Classification of National Economic Industries (GB/T 4754-2011). The adjustment of the statistical coverage has been considered.

All the nominal variables of the industrial output of China's cement industry have been adjusted based on 2000 constant prices in order to strip out the impact of intermediate inputs. The

number of employees was obtained from CEIC database, and energy consumption was obtained from China Energy Statistical Yearbooks. We chose final energy consumption instead of total energy consumption in order to calculate the CO₂ emission more accurately. The standard quantity of energy consumption was calculated by calorific value calculation. We found some abnormal values in the industry's final energy consumption. In order to make the results reliable and robust, we used the interpolation method to deal with the abnormal values.

5. Results and discussions

The results of the LMDI decomposition analysis are shown in Fig. 8. The change in the CO₂ emission from 1991 to 2010 was divided to four time periods. The choice of the time intervals is based on the policy changes over these periods. Considering that China's development plans are estimated on five-year periods, the four time intervals this paper examined corresponds to the "8th five-year" period to the "11th five-year" period.

China's cement industry has experienced steady and reasonable growth rate since the establishment of the industry. From 1949 to 1984, the average annual cement consumption per capita is 50–100 kg, which can be defined as the initial stage. From 1985, the economic take-off in China largely stimulated the cement industry with an average growth rate of 10% per annum. According to the prediction by Xu et al. [5], this trend of high increase rate will end between 2015 and 2020, after referencing the development curve of cement industries in developed countries. After that, the cement industry will enter the mature stage. The most obvious characteristic in this stage is that cement production will reduce gradually, and average annual cement consumption per capita will reach and remain at the level of about 500–700 kg. Therefore, China's cement industry will still continue to grow. As can be seen from Fig. 8, despite a modest decline in the period of 1996–2000, the overall CO₂ emission in the cement industry slowly increase and gradually stabilize, owing to increasing demand and production. The variations in carbon emissions were the results of the

Table 1
Definition of parameters and multipliers in Eq. (1).

Parameter	Description	
C	Total CO ₂ emission of China's cement industry	
E_f	Proportion of fossil fuels consumed in E	
E	Total energy consumption of China's cement industry	
Y	The industrial added value of China's cement industry	
w	Total working population of China's cement industry	
Multiplier	Abbreviation	Description
C/E_f	CI	Carbon intensity effect
E_f/E	ES	Energy structure effect
E/Y	EI	Energy intensity effect
Y/w	LP	Labor productivity effect
w	IS	Industry scale effect

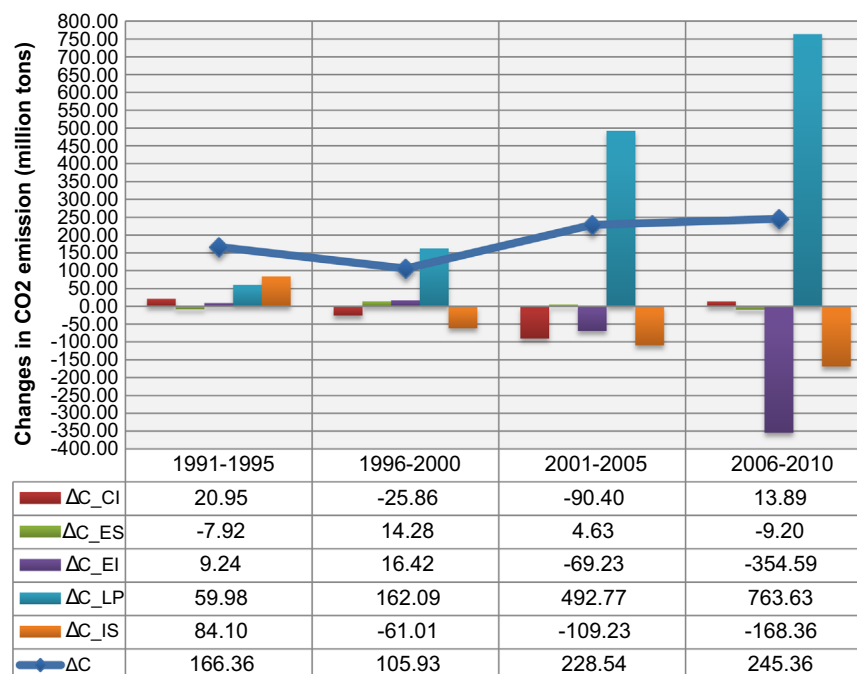


Fig. 8. Effects of driving forces for CO₂ emissions increment in China's cement industry (1991–2010).

interactions of five relevant factors: carbon intensity, energy structure, energy intensity, labor productivity and industry effect. Detailed analyses of these interactions are discussed in the following paragraphs.

5.1. First stage: 1991 to 1995

China's overall economy has achieved all-round progress since the reform and opening-up policy in year 1978. In his South Inspection Speech in 1992, Mr. Deng Xiaoping raised the opinion of "more courage and faster development". Since then, China had begun its large scale construction development, called "go all out and go fast". The rising domestic demand for cement exceeded supply, and in order to cover this gap, cement production dramatically increased in those years. However, in the period from 1991 to 1995, China's cement industry was still in an initial stage with lower industrial and technological level. Increasing the number of factories and workers are the only measures to increase total production. Therefore, although four of the five factors have positive effect, but the main contributors were: industry scale effect and labor productivity effect. In this stage of high demand, low technology and fast development, the increasing numbers of cement factories and workers were the most important driving forces of carbon emission during this period.

5.2. Second Stage: 1996 to 2000

As can be seen from Fig. 8, carbon emission in this period saw an abnormally slight decrease of about 50 million tons, although total cement production and demand had not reduced. The main reason was cement price that had increased due to the situation of excess demand in the later part of the initial period, and the increase in cement price had attracted more investments to the sector. Large amounts of newly-built cement factories caused massive progression in China's cement productivity, and then this phenomenon finally drove the cement industry into an excess supply situation. Cement price dropped significantly since then, and the industry was seen as "highly inflated". Therefore, in year 1998, a statement by the State Economic and Trade Commission (SETC) required the whole cement industry to "control production, adjust structure, close down outdated production facilities and prevent redundant construction". Alongside this statement, SETC also issued several governing policies for the purpose of "closing down small cement enterprises". Under this circumstance and combined with the financial crisis in Asia during this period, import and export businesses in China were massively affected, and large number of factories and enterprises went bankruptcy at that time. These abnormal factors reduced carbon emissions in China's cement industry while production remained on an increasing trend.

All the five factors captured the influences of this economic crisis and industrial reorganization. Closing down small factories and massive bankruptcy has huge effect in this period as the industrial scale effect dropped dramatically from positive into negative, indicating also that the number of employees working in the cement industry was largely reduced. Meanwhile, the decrease in the labor force promotes the labor productivity effect. With the surplus storage from the previous period being released into the market, labor productivity became the most significant contributor to carbon emission in this period.

5.3. Third stage: 2001 to 2005

Since the tenth "Five Year Plan", economic boost and soaring fixed asset investments stimulated cement production from 597 million tons in 2000 to 1.064 billion tons in 2005, an annual

growth rate of 12.3% (93.4 million tons). However the technological level in the cement industry had also been developed in this period, as the amount of new dry process cement increased from 60.6 million tons to 473.71 million tons, at an annual growth rate at 50.9% (82.62 million tons). Massive adoption of these kinds of new energy saving technologies in the cement industry contributed largely in reducing carbon emission in this period. As can be seen from Fig. 8, carbon intensity and energy intensity effects were both negative to carbon emissions. However, massive infrastructure constructions in this period required the output of the cement industry. There were a total of 5177 cement-related enterprises (1000 plus for cement grinding and 4000 plus for clinker production) by the end of 2005, an increase of 500 units compared to 2001. However, due to technological progression, increasing amount of cement production and cement enterprises did not increase the amount of people employed in the industry. Therefore, cement output per capita became the largest contributor to carbon emission in these five years.

5.4. Forth stage: 2006 to 2010

Carbon emissions in China's cement industry slowed down after experiencing big boosts in productivity, production, number of enterprises and carbon emissions in the previous period. Even though cement production continue to increase at a high rate, different kinds of new energy-saving technologies, which effectively controlled carbon emission, were widely employed. As shown in Fig. 8, energy intensity effect has the most significant negative effect on carbon emissions, which proves that by adopting new types of dry process methods, cogenerations and using industrial wastes as secondary fuel, energy consumption in the cement industry was reduced largely. Therefore, it offset the new carbon emission produced by growing cement production and scales. The most obvious positive force during this period was the same as in the previous period—labor productivity. Combined with large scale industrial expansion in the last period, this fact can be seen as its hysteresis effect. The increasing number of large-scale cement enterprises and the advances in technology inevitably increase labor productivity. Therefore, it became the most positive driving force of carbon emission in this period.

6. Recommendations and conclusion

Energy demand in China in the past decade had grown fastest and had huge impacts on global energy structure. In year 2002, China only consumed 12.7% of global total energy consumption, but the figure rose to 22.4% in 2011. Currently, China is the largest energy consumer and producer in the world. To sustain the current high economic growth, China needs to ensure sufficient and cheap energy supply. Therefore coal inevitably became the best choice. Currently, coal accounts for 69% of China's primary energy consumption and generates 80% of total electricity. In line with soaring energy consumption, CO₂ emission in China increased dramatically from 4.2 billion tons to 9.2 billion tons in the past ten years. Even though reducing the use of coal and developing a low-carbon economy had become a social consensus in China, the facts and figures showed the difficulty in implementing it.

Currently China is in the stage of urbanization and industrialization, which is accompanied with high economic growth rate and huge and growing energy demand. According to research results from CCEER (China Centre of Energy Economics Research), China will remain at this stage at least until year 2020. Compared with the experiences and histories of western countries, this stage of urbanization always accompany heavy-industrialization, where large-scale infrastructure constructions requires enormous

demand of outputs of high-energy-consuming industries, like the cement and iron and steel industries.

The fundamental reality in China is its huge population, and the current trend of urbanization and industrialization are accompanied by high economic growth rate and huge and growing energy demand. In year 2009, the GDP of China accounted for only 8% of total global GDP, but it consumes 46.4% of steel, 60.9% of cast iron and 48.7% of cements globally. These figures prove that urbanization is the initial driving force of China's economic growth and energy demand growth. The logical relationship is: high-speed economic growth pushed the progress of urbanization, and the urbanization progress boosted total energy consumption. Then the most obvious characteristic of urbanization and industrialization is boom in high energy-consuming industries and pollutant dischargers. Therefore, urbanization had become the crux of solving the puzzle of China's dangerous energy situation. In 2012, urbanization rate in China is 52.6%, which is still far from 61% of middle income countries and 78% of high income countries. Studies anticipated that China will slow its urbanization progress around 2020 with 300 million people migrating from rural areas to cities (almost equal to current population in US). This migration trend will increase the energy consumptions in China (according to statistical data from 1990 to 2004, the average energy consumption of urban residents are 2.8 times of rural residents).

In the meantime, huge-scale infrastructure construction requires enormous amount of cement and steels, which is too large to be imported from other countries, and hence the promotion of domestic production. The rigid demand of high energy-consuming products required by urbanization cannot be entirely offset by technological advances. Therefore, as China continue its urbanization progress, the total energy consumption and carbon dioxide emissions will still continue to increase in the coming years. Thus, challenges such as energy scarcity of energy, environmental degradation and CO₂ emissions are inevitable.

According to the global experience, within the process of urbanization, which indicates the urbanization rate of 20–70%, the economic structure transferred from agriculture to industry, and energy consumption per capita and energy intensity rose rapidly. After the urbanization has been completed, the industrial structure transferred to the tertiary industry, resulting in lower energy intensity, and energy consumption per capita slows or stabilizes. The historical experience shows that the economic development of developed countries showed the sign of heavy industry as well, indicating the need of high energy-consuming industry from massive infrastructure construction. From 2003 to 2008, total annual energy consumption increased at a double-digits rate, due to accelerated urbanization. The main growth model of China's economy is investment-oriented and high-speed growth. During the recent three decades, the average annual GDP is 9.5%. Most of the time, investment weights more than 40% of GDP, and the current level standing at about 50%. The dominator of China's economy has always been heavy industry. In 1985, the heavy industry weighted 55% of GDP. In 1990 it declined to 50%, rose back to 60% in 2000, and was as high as 69% in 2005. Under the stimulus of massive infrastructure construction due to economic growth and urbanization, the heavy industry, especially the high energy-consuming ones, experienced the most rapid growth.

When faced with the problem of energy and environment, the Chinese government would not slow urbanization at the initial point, but take it as a chance to practice energy conservation. Energy consumption will apparently display various features under different economy growth styles, energy policies and environment regulations. By planning and executing positive energy policy during urbanization, energy efficiency will be higher and the structure cleaner. Urbanization can also be seen as a process of determining lifestyles, and the lifestyle itself will

directly affect energy consumption. To advocate an energy saving lifestyle guided by policies is another important aspect of the sustainability of China's economy. To reduce energy consumption during urbanization actually involves the concept of "Low-Carbon City". The core of low-carbon economy is increasing energy efficiency and transforming the energy structure.

Exploring and establishing low-carbon economic development model in "low carbon city" could reduce carbon dioxide emissions. "Low-carbon city" includes low-emission, high energy efficiency, and high productivity as the features of planning and construction of "low-carbon city" in the process of urbanization. By adjusting the industrial structure and restructuring the development model, the low-carbon model would affect the economy at the minimum level, balance economic growth and increase employment rates. Drafting a strategy for eco-city construction plan to motivate local governments and financial companies to advocate efficient low-carbon technology of energy saving and low-emission in the cities will require policy stimulus, financial aid, technology innovation and capital flotation. Besides Beijing, Shanghai and other big cities, the second/third-tier cities would be the best places to develop low-carbon cities because of the high opportunities there.

Under the circumstance of the global financial crisis and economic slowdown, promoting green industry development through low-carbon economy could be a tool to stimulate the economy. The investments made by the Chinese government had clearly pointed out their goals of promoting industrial upgrading and integration, increasing energy efficiency and maintaining sustainable growth. The local governments should try to find new growth points within the crisis. The requirements from adjusting the industrial structure are consistent with those of developing "low-carbon city". The exploring of "low carbon city" is believed to be an achievement during the urbanization process in China. For a "low-carbon" cement industry, more energy conservative technologies should be adopted, and with the global attention on emission reduction, new researches and projects are being carried out regularly. China's cement industry should learn from those of developed countries, by using alternative fuels, especially industrial waste and biomass fuels. Compared with the proportion of 30% of less-carbon-intensive fuels in the cement industries in Germany, France and Switzerland [7], the 1% proportion in China is lagging behind. However, the transformation from conventional carbon intensive fossil fuels to environmental friendly renewable sources requires a comprehensive fundamental energy and supporting facilities. With the increasing public awareness of promoting a low-carbon earth, the Chinese government and the cement industry will have a better opportunity for its transformation.

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References

- [1] BP. Statistical Review of World Energy Full Report. 2012. <http://www.bp.com/statisticalreview>.
- [2] Chinese Ministry of Industry and Information Technology (CMIIT). Operation Status of Major Industry Sectors in 2010: Material Industry; 2011 (<http://www.miit.gov.cn/n11293472/n11295125/n11299605/13578992.html>).

- [3] U.S. Geological Survey (USGS). Mineral Commodity Summaries 2011. Washington: United States Government Printing Office; 2011.
- [4] NBS (National Bureau of Statistics). China Statistical Yearbook 2011. Beijing: China Statistics Press; 2012.
- [5] Xu J-H, Fleiter T, Eichhammer W, Fan Y. Energy consumption and CO₂ emissions in China's cement industry: a perspective from LMDI decomposition analysis. *Energy Policy* 2012;50:821–32.
- [6] IEA. World Energy Outlook 2011. (WEO 2011). Paris: OECD/IEA; 2011.
- [7] Habert G, Billard C, Rossi P, Chen C, Roussel N. Cement production technology improvement compared to factor 4 objectives. *Cem. Concr. Res.* 2010;40(5):820–6.
- [8] Chinese Ministry of Industry and Information Technology (CMIIT). Operation Status of Major Industry Sectors in 2009: Material Industry; 2010 (<http://www.miit.gov.cn/n11293472/n11293832/n12768545/13568172.html>).
- [9] IEA (International Energy Agency). Cement Technology Roadmap 2009: Carbon Emissions Reductions up to 2050. 2009. http://www.iea.org/publications/freepublications/publication/Cement_Roadmap.pdf.
- [10] Xu J-H, Fleiter T, Fan Y, Eichhammer W. CO₂ emissions reduction potential in China's cement industry compared to IEA's Cement Technology Roadmap up to 2050. *Appl. Energy* 2014;130:592–602.
- [11] Friedl B, Getzner M. Determinants of CO₂ emissions in a small open economy. *Ecol. Econ.* 2003;45(1):133–48.
- [12] Richmond AK, Kaufmann RK. Is there a turning point in the relationship between income and energy use and/or carbon emissions? *Ecol. Econ.* 2006;56(2):176–89.
- [13] Lantz V, Feng Q. Assessing income, population, and technology impacts on CO₂ emissions in Canada: where's the EKC? *Ecol. Econ.* 2006;57(2):229–38.
- [14] Wagner M. The carbon Kuznets curve: a cloudy picture emitted by bad econometrics? *Resour. Energy Econ.* 2008;30(3):388–408.
- [15] Auffhammer M, Carson RT. Forecasting the path of China's CO₂ emissions using province-level information. *J. Environ. Econ. Manag.* 2008;55(3):229–47.
- [16] Ang JB. CO₂ emissions, research and technology transfer in China. *Ecol. Econ.* 2009;68(10):2658–65.
- [17] Ang B, Zhang F, Choi K. Factorizing changes in energy and environmental indicators through decomposition. *Energy* 1998;23(6):489–95.
- [18] Liu L-C, Fan Y, Wu G, Wei Y-M. Using LMDI method to analyze the change of China's industrial CO₂ emissions from final fuel use: An empirical analysis. *Energy Policy* 2007;35(11):5892–900.
- [19] Fan Y, Liu L-C, Wu G, Tsai H-T, Wei Y-M. Changes in carbon intensity in China: Empirical findings from 1980–2003. *Ecol. Econ.* 2007;62(3–4):683–91.
- [20] Zhang M, Mu H, Ning Y, Song Y. Decomposition of energy-related CO₂ emission over 1991–2006 in China. *Ecol. Econ.* 2009;68(7):2122–8.
- [21] Sun W-q, Cai J-j, Mao H-j, Guan D-j. Change in Carbon Dioxide (CO₂) emissions from energy use in China's iron and steel industry. *J. Iron Steel Res., Int.* 2011;18(6):31–6.
- [22] Lin BQ, Wang XL. Carbon emissions from energy intensive industry in China: Evidence from the iron & steel industry. *Renewable Sustainable Energy Rev.* 2015;47:746–54.
- [23] Tian Y, Zhu Q, Geng Y. An analysis of energy-related greenhouse gas emissions in the Chinese iron and steel industry. *Energy Policy* 2013;56:352–61.
- [24] Boqiang Lin, Mohamed Moubarak. Decomposition analysis: Change of carbon dioxide emissions in the Chinese textile industry. *Renewable Sustainable Energy Rev.* 2013;26:389–96.
- [25] Zhang M, Liu X, Wang WW, Zhou M. Decomposition analysis of CO₂ emissions from electricity generation in China. *Energy Policy* 2013;52:159–65.
- [26] Ouyang XL, Lin BQ. An analysis of the driving forces of energy-related carbon dioxide emissions in China's industrial sector. *Renewable Sustainable Energy Rev.* 2015;45:838–49.
- [27] Liu FR, Marc, Wang, Shumao. Energy efficiency of China's cement industry. *Energy* 1995;20(7):669–81.
- [28] Li TP. Report on present status of energy efficiency in China's cement industry and its energy saving potentiality. *China Build. Mater.* 2004;9:23–7.
- [29] Lei Y, Zhang Q, Nielsen C, He K. An inventory of primary air pollutants and CO₂ emissions from cement production in China, 1990–2020. *Atmos. Environ.* 2011;45(1):147–54.
- [30] Worrell EGC, Price L. Energy Efficiency Improvement Opportunities for the Cement Industry. 2008.
- [31] Zeng XM. Energy consumption status and energy saving potential in cement industry. *China Cem.* 2006;3:16–21.
- [32] Zeng XM. Current status and energy efficiency improvement of China cement industry. Workshop on International Comparison of Industrial Energy Efficiency; 2008.
- [33] Hasanbeigi A, Price L, Lu H, Lan W. Analysis of energy-efficiency opportunities for the cement industry in Shandong Province, China: a case study of 16 cement plants. *Energy* 2010;35(8):3461–73.
- [34] Cui SP, Liu W. Analysis of CO₂ emission mitigation potential in cement producing processes. *China Cem.* 2008;4:57–9.
- [35] Price L, Worrell E, Sinton J, Yun J. Industrial energy efficiency policy in China. ACEEE Summer Study on Energy Efficiency in Industry in 2001. China: Lawrence Berkeley National Laboratory and China Energy Conservation Association; 2001.
- [36] Wang Y, Zhu Q, Geng Y. Trajectory and driving factors for GHG emissions in the Chinese cement industry. *J. Cleaner Prod.* 2013;53:252–60.
- [37] Teller P, Denis S, Renzoni R, Germain A, Delaisse P, D'Inverno H. Use of LCI for the decision-making of a Belgian cement producer: a common methodology for accounting CO₂ emissions related to the cement life cycle. In: The 8th LCA Case Studies Symposium SETAC-Europe; 2000.